Full results for large-sized datasets

Overview datasets

Dataset	n	d	Type	Source
Survival	110,204	4	Tabular	UCI ML Repository
Diabetes	253,680	22	Tabular	UCI ML Repository
Music	515,344	91	Tabular	Kaggle
Covtype	581,012	55	Tabular	UCI ML Repository
Imagenet8	1,281,166	192	Image	ImageNet Image Database
Imagenet32	1,281,166	3,072	Image	ImageNet Image Database
Census	2,458,284	68	Tabular	UCI ML Repository
Finance	6,362,620	12	Tabular	Kaggle

Metric: Cityblock (L1, Manhattan)

Objective function values

Dataset	n	d	Best objective	% DEV HEX-L	% DEV ABM-d1000	% DEV ABM-d2000	% DEV ABM-d3000	% DEV ABM-d4000	% DEV ABM-d5000	% DEV RND
Survival	110,204	4	271,069.32	1.35	0.02	0.00	0.00	0.00	0.00	31.31
Diabetes	253,680	22	3,179,814.96	20.06	0.03	0.00	0.00	0.00	0.00	29.13
Music	515,344	91	28,304,179.67	16.94	1.25	0.68	0.37	0.16	0.00	17.44
Covtype	581,012	55	9,031,204.35	21.02	0.28	0.11	0.05	0.02	0.00	22.10
Imagenet8	1,281,166	192	44,254,178.48	26.62	1.25	0.66	0.35	0.15	0.00	26.68
Imagenet32	1,281,166	3,072	770,120,739.88	—	1.28	0.68	0.36	0.16	0.00	24.22
Census	2,458,284	68	85,474,467.03	26.45	0.21	0.09	0.04	0.02	0.00	26.47
Finance	6,362,620	12	24,770,280.56	22.36	0.01	0.00	0.00	0.00	0.00	22.40
Average				—	0.54	0.28	0.15	0.06	0.00	24.97

Running times (seconds)

Dataset	n	d	CPU HEX-L	CPU ABM-d1000	CPU ABM-d2000	CPU ABM-d3000	CPU ABM-d4000	CPU ABM-d5000	CPU RND
Survival	110,204	4	7,203.67	0.51	1.56	3.90	5.45	9.00	0.01
Diabetes	253,680	22	7,241.69	1.74	6.79	16.83	29.31	49.66	0.02
Music	515,344	91	7,519.63	3.50	11.34	22.69	37.41	56.33	0.04
Covtype	581,012	55	7,416.01	4.74	18.31	39.76	75.27	122.31	0.04
Imagenet8	1,281,166	192	8,858.97	9.48	31.95	65.73	112.75	161.52	0.10
Imagenet32	1,281,166	3,072	—	61.27	134.53	223.05	308.88	407.88	0.10
Census	2,458,284	68	8,355.82	25.03	102.06	224.31	409.41	663.67	0.19
Finance	6,362,620	12	7,969.95	67.57	276.17	606.69	1,145.63	1,761.84	0.49
Average			—	21.73	72.84	150.37	265.51	404.03	0.12

Metric: Euclidean (L2)

Objective function values

Dataset	n	d	Best objective	% DEV HEX-L	% DEV ABM-d1000	% DEV ABM-d2000	% DEV ABM-d3000	% DEV ABM-d4000	% DEV ABM-d5000	% DEV RND
Survival	110,204	4	184,106.25	3.31	0.02	0.01	0.00	0.00	0.00	25.48
Diabetes	253,680	22	995,618.11	12.66	0.52	0.25	0.13	0.05	0.00	18.78
Music	515,344	91	3,743,627.11	14.99	1.01	0.54	0.29	0.12	0.00	15.47
Covtype	581,012	55	3,065,968.79	11.56	0.31	0.15	0.08	0.03	0.00	12.31
Imagenet8	1,281,166	192	3,660,509.02	23.46	1.11	0.59	0.31	0.13	0.00	23.48
Imagenet32	1,281,166	3,072	15,942,857.72	—	1.09	0.58	0.31	0.13	0.00	20.61
Census	2,458,284	68	16,300,088.82	15.55	0.24	0.11	0.06	0.02	0.00	15.59
Finance	6,362,620	12	13,553,068.41	20.01	0.00	0.00	0.00	0.00	0.00	20.04
Average				—	0.54	0.28	0.15	0.06	0.00	18.97

Running times (seconds)

Dataset	n	d	CPU HEX-L	CPU ABM-d1000	CPU ABM-d2000	CPU ABM-d3000	CPU ABM-d4000	CPU ABM-d5000	CPU RND
Survival	110,204	4	7,203.36	0.82	3.08	8.85	12.67	21.39	0.01
Diabetes	253,680	22	7,241.47	1.79	6.56	15.47	26.49	43.36	0.02
Music	515,344	91	7,518.42	3.84	12.81	24.84	42.18	65.96	0.04
Covtype	581,012	55	7,414.50	5.85	22.71	47.45	85.68	133.88	0.04
Imagenet8	1,281,166	192	8,883.62	10.06	34.36	69.57	122.50	179.50	0.10
Imagenet32	1,281,166	3,072	—	58.77	132.63	221.03	312.60	416.64	0.10
Census	2,458,284	68	8,351.24	26.67	108.01	237.42	435.50	700.26	0.20
Finance	6,362,620	12	7,973.17	71.82	279.02	610.52	1,151.63	1,770.96	0.50
Average			—	22.45	74.90	154.39	273.66	416.49	0.13

Metric: Squared Euclidean (L2 squared)

Objective function values

Dataset	n	d	Best objective	% DEV HEX-L	% DEV ABM-d1000	% DEV ABM-d2000	% DEV ABM-d3000	% DEV ABM-d4000	% DEV ABM-d5000	% DEV FAC-5	% DEV RND
Survival	110,204	4	663,840.25	2.43	0.03	0.01	0.00	0.00	0.00	4.60	33.53
Diabetes	253,680	22	7,979,938.03	20.31	0.73	0.33	0.16	0.06	0.00	—	30.05
Music	515,344	91	61,814,684.86	23.34	2.02	1.10	0.59	0.25	0.00	—	24.15
Covtype	581,012	55	37,666,117.82	14.24	0.41	0.20	0.10	0.04	0.00	—	15.16
Imagenet8	1,281,166	192	23,109,921.67	42.13	2.28	1.21	0.65	0.27	0.00	—	42.36
Imagenet32	1,281,166	3,072	428,597,408.33	—	2.24	1.19	0.64	0.27	0.00	—	38.01
Census	2,458,284	68	219,878,062.39	23.92	0.32	0.14	0.07	0.03	0.00	—	23.98
Finance	6,362,620	12	88,425,514.07	13.59	0.00	0.00	0.00	0.00	0.00	—	13.65
Average				—	1.00	0.52	0.28	0.12	0.00	—	27.61

Running times (seconds)

Dataset	n	d	CPU HEX-L	CPU ABM-d1000	CPU ABM-d2000	CPU ABM-d3000	CPU ABM-d4000	CPU ABM-d5000	CPU FAC-5	CPU RND
Survival	110,204	4	7,204.08	0.66	2.23	5.86	8.13	13.94	2.89	0.01
Diabetes	253,680	22	7,241.48	1.69	6.24	14.58	24.85	41.67	—	0.02
Music	515,344	91	7,524.23	3.35	10.92	21.66	34.61	53.22	—	0.04
Covtype	581,012	55	7,417.57	5.49	21.34	43.84	81.94	126.70	—	0.04
Imagenet8	1,281,166	192	8,869.02	8.47	27.97	55.34	95.51	143.84	—	0.10
Imagenet32	1,281,166	3,072	—	58.02	127.72	210.64	292.38	386.26	—	0.10
Census	2,458,284	68	8,371.66	26.55	109.67	241.25	435.05	702.47	—	0.19
Finance	6,362,620	12	7,983.34	70.86	283.36	616.68	1,157.23	1,773.48	—	0.49
Average			—	21.89	73.68	151.23	266.21	405.20	—	0.12